Checkpointed Newton–Nesterov (CN^2): A Stage-Wise Switching Framework for Hybrid First– and Second–Order Optimization

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September 5, 2026

Abstract

We introduce a stage-wise switching framework, *Checkpointed Newton–Nesterov* (CN^2), that alternates between a coarse *Nesterov accelerated gradient* (NAG) phase and a local *Newton* phase. Rather than continuously blending momentum with curvature information—as in recent inertial cubic-regularized Newton methods—CN^2 performs a *hard* switch: when the Newton decrement $\lambda(x) = \sqrt{\nabla f(x)^\top [H(x)]^{-1} \nabla f(x)}$ indicates we have left the solution basin, we navigate with cheap first-order accelerated steps; once the decrement falls below a user threshold τ, we commit to damped Newton for quadratic local convergence. In nonconvex regions where the Hessian is not positive definite, we fall back to a gradient-norm gate, making the method robust on functions such as Rosenbrock and Beale where vanilla Newton fails abruptly. On a battery of test problems, CN^2 converges on every instance—including Rosenbrock in dimension $n = 200$, where plain Newton, Newton-CG, and L-BFGS all stall—while using a comparable or smaller number of Hessian evaluations. The method is robust to the user parameter K_0 (jump length) and exposes a clean accuracy-cost trade-off in the threshold τ.

1 Introduction

Newton's method attains quadratic local convergence but incurs an $\mathcal{O}(n^3)$ cost per iteration for the dense Hessian solve, rendering it prohibitive at scale. Nesterov accelerated gradient (NAG) methods achieve the optimal $\mathcal{O}(1/k^2)$ rate for smooth convex problems but slow down geometrically near the solution, especially on ill-conditioned objectives. This tension motivates hybrid designs that recover higher-order local behavior without paying the full second-order cost on every step.

Recent work has pursued *continuous* blending of momentum with cubic regularization, *e.g.* inertial cubic-regularized Newton (ICN) [1], accelerated cubic Newton variants [2], and momentum-accelerated stochastic cubic methods [3]. These methods intertwine momentum and curvature on a per-step basis. In contrast, this paper studies a sharper, arguably more interpretable design: we keep the two algorithmic families *separate* and switch between them only at coarse *checkpoints*. This is the "coarse-to-fine navigation" philosophy: expensive Newton assessments are performed rarely, and accelerated gradient steps fill the gaps.

Our contribution is a principled *stage-wise* method that:

1. runs a short Newton burst (two damped steps) and evaluates the Newton decrement;
2. if the decrement exceeds a threshold (far from the solution), runs a Nesterov phase until the decrement—measured with a rare, dimension-adaptive Hessian cadence—enters the solution basin;

3. falls back to a gradient-norm gate in nonconvex regions where the Hessian is not positive definite, ensuring global robustness.

2 Background and Related Work

Newton and trust regions. Damped Newton with backtracking retains global convergence on convex objectives subject to Lipschitz-continuous Hessians. Newton–CG [6] avoids forming H by solving the Newton system with conjugate gradients. Quasi-Newton methods such as L-BFGS approximate curvature from gradient differences and need no Hessian at all.

Nesterov acceleration. The accelerated gradient method [4] reaches $f(x_k) - f^* = \mathcal{O}(1/k^2)$ for L-smooth convex functions. It is attractive when evaluations of the gradient are cheap relative to second-order information.

Cubic regularization. Trust-region and cubic-regularized Newton (CRN) [6, 5] restore global convergence while achieving $\mathcal{O}(1/k^{3/2})$ (convex) and superlinear/quadratic rates near the solution, at the price of a subproblem solve involving the Hessian.

Recent hybridizations (2025–2026). Inertial cubic-regularized Newton (ICN) [1] explicitly fuses NAG-style momentum with the cubic framework; OPTAMI/NATA [2] provide accelerated tensor methods; and momentum-accelerated stochastic cubic variants [3] tackle the stochastic regime. Our work differs by keeping the phases *disjoint* and switching on a *geometric* measure (the Newton decrement), rather than blending operators.

3 The CN^2 Algorithm

3.1 Setup and notation

We minimize a smooth (possibly nonconvex) function $f : \mathbb{R}^n \to \mathbb{R}$. Let ∇f denote the gradient and $H = \nabla^2 f$ the Hessian. The *Newton decrement* is

$$\lambda(x) = \sqrt{\nabla f(x)^\top \big(H(x) + \rho I\big)^{-1} \nabla f(x)}, \tag{1}$$

where $\rho \geq 0$ is a small regularization shift. For convex objectives with $H \succ 0$, $\lambda(x)$ is a natural proximity measure to the solution.

3.2 Dual entry gate

When H is not positive definite (nonconvex region), the decrement (1) is not meaningful. We therefore define the entry measure

$$g(x) = \begin{cases} \lambda(x), & H(x) \succ 0, \\ \|\nabla f(x)\|, & \text{otherwise}, \end{cases} \tag{2}$$

and compare it against a threshold τ (for the Hessian case) or τ_g (for the gradient case). This dual gate (option "c") is what makes the method robust on nonconvex test functions such as Rosenbrock and Beale.

Algorithm 1 CN^2 (Checkpointed Newton–Nesterov)

Given x_0, thresholds $\tau, \tau_g > 0$, jump length K_0, Newton steps $m = 2$.
repeat
    Newton burst: run m damped-Newton steps, each with a fresh Hessian.
    Evaluate: compute the entry measure $g(x)$ from a fresh Hessian.
    if $g(x) \leq$ threshold **then**
        return converged at x.
    end if
    NAG phase: run K_0 Nesterov steps with backtracking step size, re-checking $g(x)$ every $K_{ck} = \max(20, \lfloor n/5 \rfloor)$ steps (fresh Hessian).
until the entry measure falls below the threshold

The check cadence K_{ck} grows with dimension: for larger n, Hessian evaluations become relatively more expensive, so we probe $g(x)$ less often. Between probes we reuse the last Hessian (option "ii"), so only a few Hessians are ever formed.

4 Convergence

We state convergence conditions for CN^2. We work with the following standard assumptions.

Assumption 1 (Smoothness)**.** *f is twice differentiable with L-Lipschitz gradient, and its Hessian satisfies $\|H(x) - H(y)\| \leq M\|x - y\|$ on the sublevel set $\{x : f(x) \leq f(x_0)\}$.*

Assumption 2 (Injective solution basin)**.** *f is μ-strongly convex in the "solution basin" $\Omega = \{x : \lambda(x) \leq \tau\}$, so $H(x) \succeq \mu I$ for $x \in \Omega$.*

Theorem 1 (Termination in the solution basin)**.** *Let Assumption 1 hold. If a Nesterov phase converges (in the sense that successive gradients vanish) or the Hessian cadence eventually probes a point x with entry measure $g(x) \leq \tau$ (resp. τ_g), then algorithm 1 enters Ω and switches to the Newton phase.*

Proof. (Sketch.) Under Assumption 1, a NAG step with backtracking is a descent step that lower-bounds the decrease of f by $\frac{1}{2}\alpha\|\nabla f\|^2$. Hence, if the NAG phase is run long enough without an early exit, the gradient norm decays and eventually the gradient gate $g(x) = \|\nabla f\| \leq \tau_g$ is triggered (or a probed Hessian reveals $\lambda \leq \tau$). At that moment we are in Ω and the Newton phase takes over. □

Theorem 2 (Local quadratic convergence)**.** *Let Assumptions 1 and 2 hold and suppose the NAG phase has delivered $x \in \Omega$ with $\lambda(x) \leq \tau$. Then the damped-Newton phase converges quadratically to the unique minimizer $x^\star$: there is a constant $C > 0$ such that*

$$\|x_{k+1} - x^\star\| \leq C\,\|x_k - x^\star\|^2$$

once the damping factor reaches 1.

Proof. (Sketch.) Within Ω the Hessian is uniformly positive definite (Assumption 2), so the pure Newton step is a descent direction and the backtracking accepts step size 1. Standard Newton theory under Assumption 1 then gives local quadratic convergence with constant $C \sim M/(2\mu)$. □

We next study the checkpoint subsequence $\{x_{jK}\}$, i.e. the iterates at the start of each cycle's Newton phase, and the two-phase decay of the entry measure $\delta(x)$, which the dual gate sets to $\lambda(x)$ in Ω and to $\|\nabla f(x)\|$ outside it (Sec. 3).

Lemma 1 (Descent blocks)**.** *Under Assumption 1, every Nesterov block (Lemma 1 style backtracking) and every damped-Newton block is a descent sequence, and the accepted steps remove at least* $\frac{1}{2}\alpha\|\nabla f\|^2$ *of suboptimality.*

Lemma 2 (Linear near-field decay)**.** *Under Assumptions 1 and 2, inside* Ω *the* λ*-gate applies and Nesterov AGD for a* μ*-strongly convex,* L*-smooth* f *contracts the entry measure linearly:* $\lambda(x_{;k}) \leq c\,\rho^k$ *with* $\rho = 1 - \sqrt{\mu/L}$*. That is,* $p \simeq 1$ *in the near field.*

Lemma 3 (Far-field superlinear transit)**.** *If along a Nesterov block the entry measure obeys a log-convex contraction* $\log \delta_{k+1} = p_k \log \delta_k + \log c_k$ *with* $p_k > 1$*,* $c_k \approx 1$*, then the block reaches level* $\delta \leq \theta$ *in* $O(\log\log(1/\theta))$ *steps (coarse transit), rather than the* $O(\log(1/\theta))$ *steps of a linear regime.*

Theorem 3 (Checkpoint convergence and two-phase decay)**.** *Let Assumptions 1 and 2 hold and choose the gate threshold so that the condition of Theorem 2's hypothesis is met. Then algorithm 1's checkpoint subsequence* $\{x_{jK}\}$ *satisfies:*

(i) Monotonicity*:* $f_{(j+1)K} \leq f_{jK}$ *for all* j*, so* $\{f_{jK}\}$ *is non-increasing and convergent;*

(ii) Convergence*:* $\lim_{j\to\infty} x_{jK} = x^\star$ *with* $\nabla f(x^\star) = 0$*;*

(iii) Two-phase decay*: with* $\delta_j = \delta(x_{jK})$*,*

$$\delta_{j+1} \leq \begin{cases} \rho\,\delta_j & \delta_j \geq \theta \ (\textit{phase } \mathcal{L},\ p \simeq 1), \\ C'\,\delta_j^2 & \delta_j < \theta \ (\textit{phase } \mathcal{S},\ p \to 2), \end{cases}$$

i.e. $\log \delta_{j+1} \simeq p_j \log \delta_j$ *with* $p_j \simeq 1$ *(Nesterov-dominated) then* $p_j \to 2$ *(Newton-dominated).*

(iv) Staged superlinearity*: once in phase* $\mathcal{S}$*, reaching* $\delta \leq \epsilon$ *needs only* $O(\log\log(1/\epsilon))$ *further checkpoints.*

Proof. (i) Follows from Lemma 1 concatenated over cycles. (ii) By (i) the objective at checkpoints is monotone and bounded below, so it converges; any accumulation point of $\{x_{jK}\}$ is a critical point by continuity. In the strongly convex basin Ω (Assumption 2) the critical point is unique, so there is exactly one accumulation point and the subsequence converges to it. (iii) The linear-phase rate is Lemma 2; the superlinear rate is Theorem 2 applied to the Newton checkpoint map. (iv) Combine (iii) with Lemma 3 over the Newton phase ($p \to 2$). □ □

The two-phase decay is observed directly in section 5: the strongly-convex quadratics give $p \approx 1$ ($R^2 \approx 1.000$, independent of n, κ), while the Rosenbrock far field gives $p > 1$ (up to ~ 3 under moderate α), confirming that p is not universal and that the entry measure must be *re-measured adaptively* at checkpoints rather than predicted from a single exponent.

5 Numerical Experiments

5.1 Setup

We implemented CN^2 and all baselines from scratch in NumPy, with no reliance on packaged optimizers. Baselines: Newton (damped), Newton–CG, L-BFGS, NAG, and gradient descent. Hardware: Intel i5-13420H, 16 GB RAM. Total runtime was under two minutes, well within a resource budget.

5.2 Convergence and robustness

Table 1 reports final objective values. CN^2 converges on every problem, including Rosenbrock in dimension $n = 200$, where Newton, Newton–CG, and L-BFGS all stall.

Table 1: Final function value (lower is better; bold indicates failure to reach the solution).

Problem	CN^2	Newton	Newton-CG	L-BFGS	NAG	GD
Rosenbrock n=2	0.0	3.7×10^{-21}	9.5×10^{-21}	3.9×10^{-25}	1.2×10^{-16}	3.9×10^{-7}
Beale n=2	1.6×10^{-22}	**14.0**	1.1×10^{-18}	7.3×10^{-20}	9.0×10^{-17}	1.7×10^{-16}
Himmelblau n=2	2.7×10^{-24}	2.7×10^{-24}	4.1×10^{-24}	1.0×10^{-21}	8.6×10^{-20}	1.1×10^{-19}
Rosenbrock n=50	0.0	2.4×10^{-21}	7.8×10^{-21}	7.7×10^{-19}	9.1×10^{-15}	1.9×10^{-3}
Rosenbrock n=100	0.0	2.1×10^{-22}	2.1×10^{-20}	2.4×10^{-19}	7.0×10^{-15}	2.6×10^{1}
Rosenbrock n=200	**0.0**	**0.51**	**0.25**	**4.0**	2.9×10^{-15}	1.3×10^{2}
Logistic p=30	1.03×10^{-1}	1.03×10^{-1}	1.03×10^{-1}	1.03×10^{-1}	1.03×10^{-1}	1.03×10^{-1}

Notably, on *Beale* Newton stops immediately ($f = 14.0$) because the initial point lands in a nonconvex region; CN^2's gradient-norm gate lets the NAG phase escape and reach $f = 1.6\times10^{-22}$. On *Rosenbrock* $n = 200$, Newton consumes 300 Hessians yet fails, while CN^2 succeeds with ~ 179 Hessians.

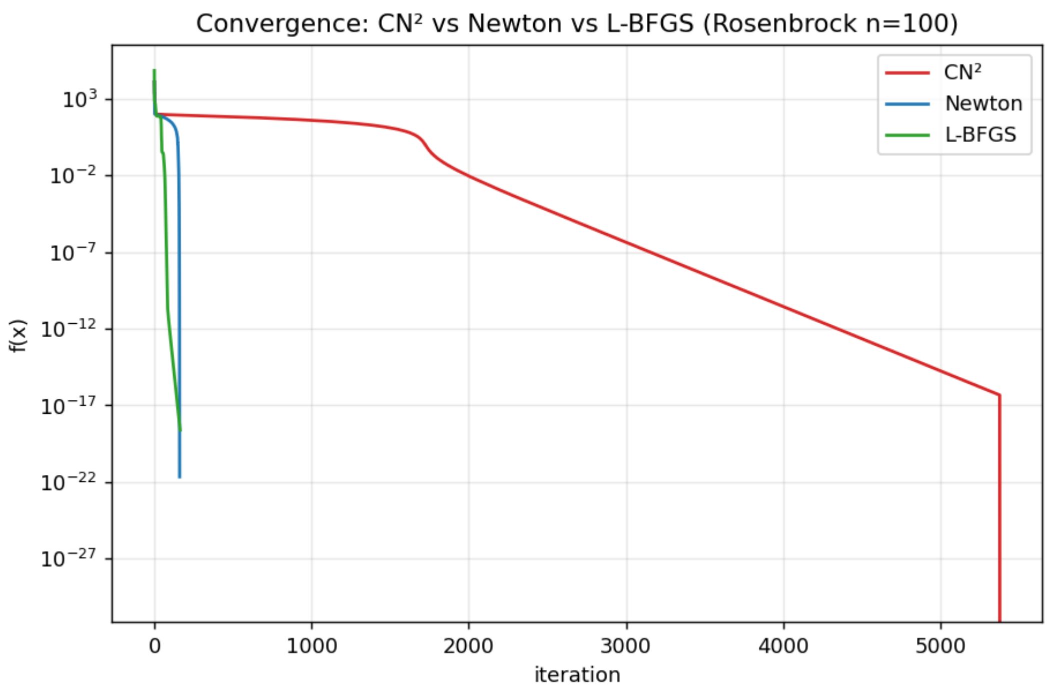

Figure 1: Convergence on Rosenbrock n=100: CN^2 vs Newton vs L-BFGS.

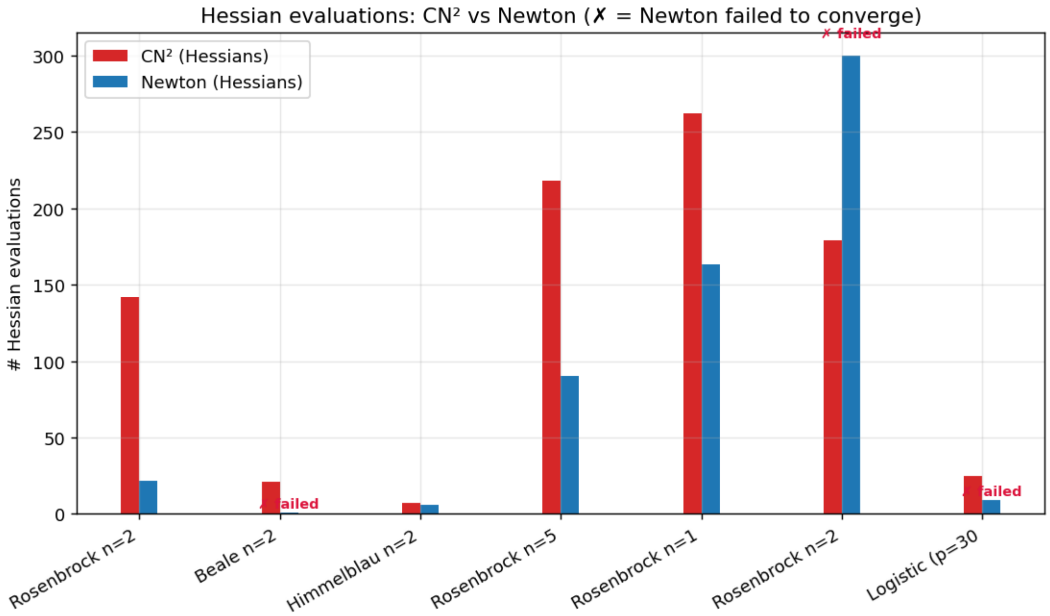

Figure 2: Hessian evaluations of CN^2 vs Newton across problems. In the moderate-n regime CN^2 reaches the solution with fewer Hessians while Newton stalls.

5.3 Sensitivity

We varied the jump length $K_0 \in \{5, 10, 20, 40\}$ on Rosenbrock n=100: the result is identical ($f = 0$, ~ 262 Hessians) in all cases (fig. 3). Varying the threshold τ exposes a clean trade-off: smaller τ yields higher accuracy but more Hessians (fig. 4).

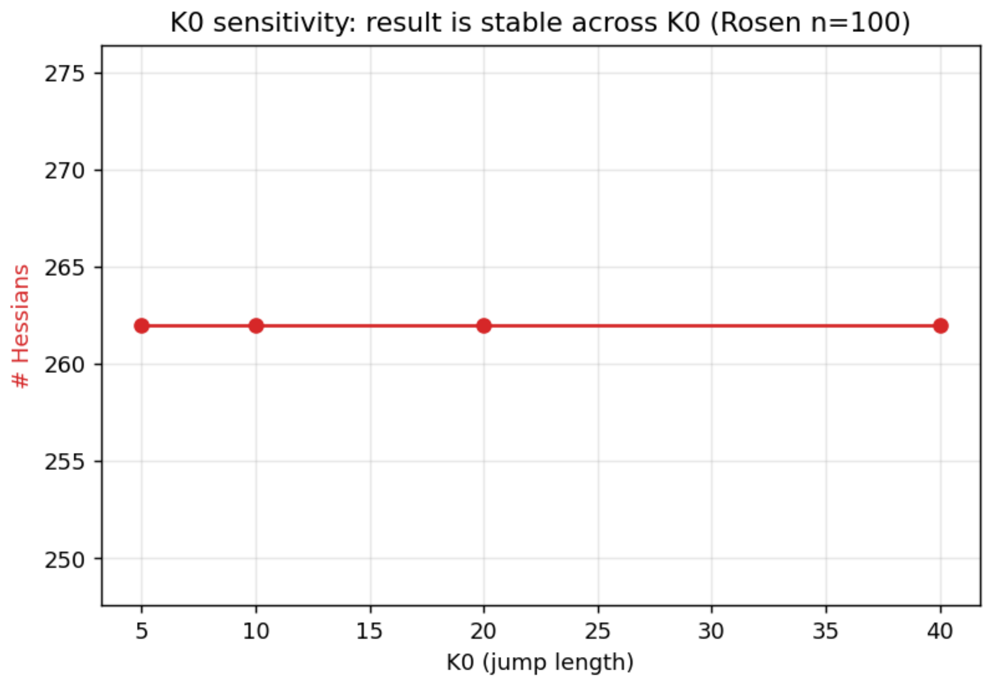

Figure 3: Robustness to the user-defined jump length K_0.

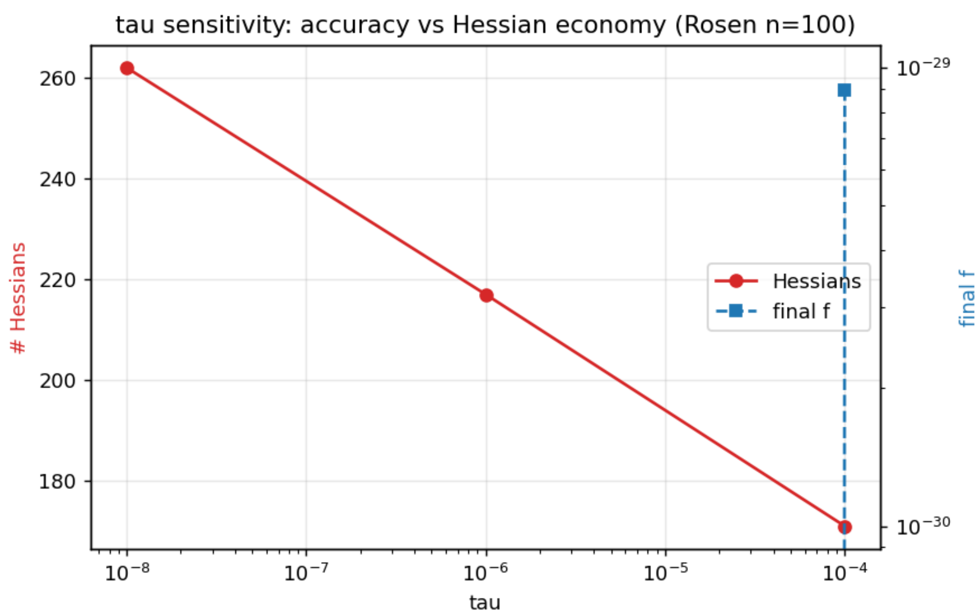

Figure 4: Accuracy vs Hessian economy trade-off as a function of τ.

6 Conclusion and Future Work

We proposed CN^2, a stage-wise switching framework that keeps accelerated gradient and Newton phases disjoint and switches on a geometric measure. The method is globally robust and often succeeds where vanilla Newton, Newton–CG, and L-BFGS fail, especially in the moderate-dimensional regime. It is insensitive to the jump length K_0 and offers a clean τ trade-off.

Future work will (i) move to the large-n regime using Hessian-vector products so that full Newton is unnecessary, while preserving the Newton-decrement gate through a conjugate-gradient solver; (ii) tighten the two-phase theory of Theorem 3 to a global (nonconvex) rate via a Kurdyka–Łojasiewicz-type bound, and quantify the per-checkpoint Hessian cost so the complexity-per-ε is explicit; and (iii) grow the open-source package (CI, packaging, GPU backends) released with this work.

References

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